MIL-M-20482D (MU)
Amendment 4
12 December 1968

MILITARY SPECIFICATION

MOTOR, ROCKET, M95
PARTS FOR

This amendment forms a part of Military Specification MIL-M-20482D (MU) dated 26 February 1965.

Page 1, Para. 2.1, add under Military Specifications:

"MIL-I-6866 - Inspections, Penetrant Method of.
MIL-I-6868 - Inspection Process, Magnetic Particle."

*Page 2, Para. 2.1, add under Standards:

"MIL-STD-1168 - Lot Numbering of Ammunition."

*Page 2, Para. 2.1, under Publications:

Change "IEL-8800014 - Index of Inspection Equipment Lists"

To " ET-8800014 - Equipment Tabulation."

Page 3, Para. 3.4 change paragraph to read as follows:

"3.4 Strength of Adapter Weld. - The adapter weld in the motor body assembly shall withstand application of a tensile force of 6,000 pounds, minimum, with no evidence of weld failure as determined by magnetic particle or penetrant inspection."

*Page 3, add 3.6 as follows:

"3.6 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract."

*Page 3, change Para. 4.1 and sub-paragraphs to read as follows:

"4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except

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as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization."

*Page 5, Para. 4.2 and all sub-paragraphs, change Initial production inspection to read First Article Inspection wherever referenced.

Page 5, Paragraph 4.2.1, delete the last sentence.

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*Page 6, change Para. 4.3.1 to read as follows:

"4.3.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist privided the lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, specification or revision thereof. Changes to either the process, specification or drawing not affecting safety, performance, interchangeability, or storage as determined by the Government shall not be deemed to alter the homogeneity of the lot. Inspection lots shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor."

Page 11, Para. 4.3.2.8, change Major defect 105. to read as follows:

"105. Straightness of motor body.....Gage 08005"

Page 11 Para. 4.3.2.8 add Major defect 107:

"107. Concentricity of inside diameter with outside diameter....Gage 08017"

*Page 14, Para. 4.3.3.1, change Initial Production to First Article wherever referenced.

*Page 16, change Para. 4.3.4 and all subparagraphs to read as follows:

"4.3.4 Inspection Equipment. - Equipment Tabulation Number ET-8800014 identifies the inspection equipment required to perform the examination and test prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

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4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, contruction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed in the ET shall be in accordance with MIL-I-45607."

Page 17, Para. 4.4.2, line 6, after magnetic particle method, add:

"or the penetrant method."

After evidence of failure, add:

"If the magnetic particle method is used, examination shall be conducted in accordance with MIL-I-6868. If the penetrant method is used, examinations shall be in accordance with MIL-I-6866.

Page 17, Para. 4.4.3, line 16, delete:

"diameter before motor body threads."

Insert in its place:

"counterbore."

Page 18, Para. 4.4.5, change first sentence to read as follows:

"This test shall be conducted in accordance with Federal Test Method Standard 151, Method 211.1 using a T2 specimen."

*Page 19, Delete Para. 5.3 in its entirety.

*Page 19, change Para. 6. and all subparagraphs to read as follows:

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

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6.2 Inspection code numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for ewhich the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required . Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

*Page 22 Para. 6.4, change initial production to first article wherever referenced.

*Asterisks indicate changes or additions to Amendment 3.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1340

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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